Test Series No.

1.10

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 14/03/2026

SYLLABUS

Physics:	11th Full Syllabus
Chemistry:	11th Full Syllabus
Botany:	11th Full Syllabus
Zoology:	11th Full Syllabus

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

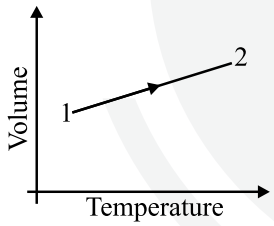
1. A quantity f is given by $f=\sqrt{\frac{hc^5}{G}}$ where c is speed of light, G universal gravitational constant and h is the Planck's constant. Dimension of f is same as that of:

(1) energy
(2) momentum
(3) area
(4) volume

2. Two spheres of masses m and $2m$ with radii r and $2r$ respectively are released in free space with a distance of $9r$ between their centres. If they move solely due to mutual gravitational attraction, the distance moved by the smaller sphere (m) just before they collide is:

(1) $2r$ (2) $4r$
(3) $3r$ (4) $6r$

3. On a volume-temperature diagram for an ideal gas, a process 1-2 is an upward sloping straight line having the tendency to cut the volume axis as shown in the figure ahead. During this process the pressure

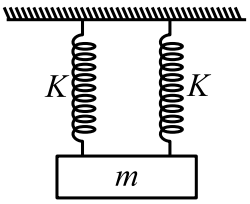


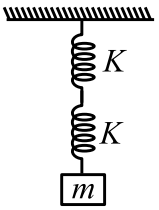
(1) Remains constant
(2) Continuously increases
(3) Continuously decreases
(4) Data Insufficient

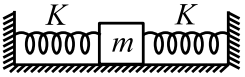
4. A metal ring has a diameter of 10.00 cm at 20°C. To fit it onto a wheel of diameter 10.02 cm (at 20°C), to what temperature must the ring be heated? (Coefficient of linear expansion of metal ring is $\alpha=2\times10^{-5}/°C$ and assume no expansion of wheel)

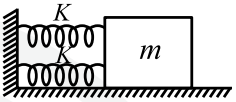
(1) 100°C (2) 220°C
(3) 120°C (4) 80°C

5. A mass m is attached to two springs of same spring constant K, as shown in following four different arrangements. In which of the following arrangements is the time period of oscillation of the mass maximum?

(1) 

(2) 

(3) 

(4) 

6. Let I_S be the moment of inertia of a solid sphere about its diameter and I_D be the moment of inertia of a solid disc about its central axis perpendicular to its plane. Both have the same mass M and radius R. If $5I_S=xI_D$, the value of x is:

(1) 2 (2) 4
(3) 1 (4) 5

7. A missile is launched from Earth's surface with a velocity less than the escape velocity. The sum of its kinetic and potential energy is:

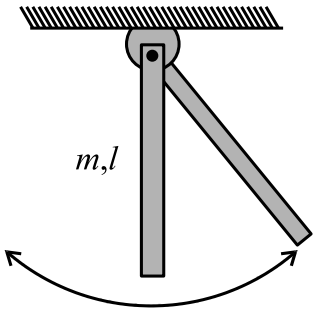
(1) positive
(2) negative
(3) zero
(4) Data insufficient

8. Two discs rotate about their respective axis with angular speed 6 rad/s and 3 rad/s in the same direction (sense). The moment of inertia of disc-1 is $I_1=2\ \text{kg m}^2$, and of disc-2 is $I_2=4\ \text{kg m}^2$. They are brought together and start rotating coaxially as a single system. The final kinetic energy of the system is:

(1) 96 J
(2) 48 J
(3) 24 J
(4) 72 J

[Contd...

9. A uniform rod of mass m and length l is suspended about its end as shown. Time period of small angular oscillations is: (g is acceleration due to gravity)



(1) $2\pi\sqrt{\frac{l}{g}}$

(2) $2\pi\sqrt{\frac{2l}{g}}$

(3) $2\pi\sqrt{\frac{2l}{3g}}$

(4) $2\pi\sqrt{\frac{l}{3g}}$

10. A 2 kg object moving along positive x-axis at 20 m/s enters a region with a retarding force $F=(-2x)$ N (where x is in m). What is its final kinetic energy after traveling from $x=0$ to $x=5$ m?

(1) 375 J (2) 400 J

(3) 425 J (4) 350 J

11. In a region of space, the gravitational potential is given by the function $V=-(2x+2y+2z)$ J/kg. The gravitational field intensity $\vec{I}$ at the point $(1,1,1)$ m is:

(1) $2\left(\hat{i}+\hat{j}+\hat{k}\right)$ N/kg

(2) $-\left(\hat{i}+\hat{j}+\hat{k}\right)$ N/kg

(3) Zero

(4) $4\left(\hat{i}+\hat{j}+\hat{k}\right)$ N/kg

12. An object falls freely from rest. If D_1, D_2, and D_3 are the distances covered in the first 2 seconds, the next 2 seconds, and the subsequent 2 seconds respectively, then:

(1) $D_1=D_2=D_3$

(2) $D_1:D_2:D_3=1:2:3$

(3) $D_1:D_2:D_3=1:3:5$

(4) $D_1:D_2:D_3=1:4:9$

13. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The center of mass of a closed loop made of a uniform thin wire, consisting of a semicircular arc of radius R and a straight wire along its diameter, lies at a distance of $\frac{2R}{\pi+2}$ from the center of the diameter.

Reason (R): For a composite system of uniform rigid bodies, the center of mass is the mass-weighted average (sum of mass × position divided by total mass) of the individual centres of mass of its constituent parts.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

14. A wheel of moment of inertia 1 kg m^2 and radius 0.5 m is rotating about its axis with a linear speed of its rim as 10 m/s. If it is brought to rest by a brake in 5 seconds, the magnitude of the average torque applied is:

(1) 2 Nm

(2) 10 Nm

(3) 4 Nm

(4) 8 Nm

15. A communication satellite orbits the Earth at a distance r from centre of the Earth. Its time period is T. If another satellite orbits at a distance $4r$ from the center of the Earth, its time period will be:

(1) $4T$ (2) $8T$

(3) $2\sqrt{2}T$ (4) $16T$

16. A heavy crate of weight 500 N is suspended vertically from a rigid support by a thick metal chain of mass 10 kg. If $g=10$ m/s^2, the force exerted by the support on the top end of the chain is:

(1) 500 N

(2) 100 N

(3) 600 N

(4) 510 N

[Contd...

17. A given sample of an ideal gas occupies a volume V at an absolute pressure P and absolute temperature T. The mass of each molecule of the gas is m. Which of the following gives the density of the gas? (k is Boltzmann constant)

(1) $\frac{P}{(kT)}$

(2) $\frac{Pm}{(kT)}$

(3) $\frac{P}{(kTV)}$

(4) mkT

18. A circular disc has a radius of 1.5 m. The area of the disc, calculated with the correct number of significant figures, is:

(1) 7.07 m^2 (2) 7.1 m^2

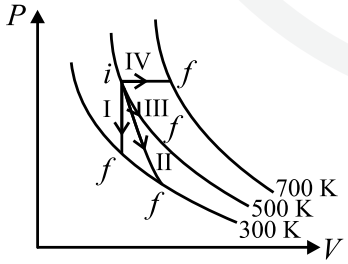
(3) 7.068 m^2 (4) 7 m^2

19. A 10 kg block is placed on a rough horizontal surface. A horizontal force of 60 N is applied, causing the block to slide 100 m in 10 s starting from rest. The coefficient of kinetic friction μ_k is: (Take $g = 10\ m/s^2$)

(1) 0.5 (2) 0.2

(3) 0.6 (4) 0.4

20. Thermodynamic processes are indicated in the following (P-V) diagram from an initial state i to a final state f). Match the following.



List-I		List-II	
(A)	Process I	(I)	Adiabatic
(B)	Process II	(II)	Isobaric
(C)	Process III	(III)	Isochoric
(D)	Process IV	(IV)	Isothermal

Choose the **correct** answer from the options given below:

(1) A-I, B-III, C-IV, D-II

(2) A-III, B-I, C-IV, D-II

(3) A-III, B-IV, C-II, D-I

(4) A-IV, B-II, C-I, D-III

21. A ball is thrown vertically downward from a height of 5 m with initial speed v_0. It hits the ground, loses 50% of its total energy, and rebounds to the same height of 5 m. The value of v_0 is: (Take $g = 10\ m/s^2$) (neglect air resistance)

(1) 5 m/s (2) 10 m/s

(3) 15 m/s (4) 20 m/s

22. A Vernier Caliper has a main scale where 1 division equals 1 mm. If 19 divisions of the main scale coincide exactly with 20 divisions of the vernier scale, the least count of the instrument is:

(1) 0.05 mm

(2) 0.01 mm

(3) 0.1 mm

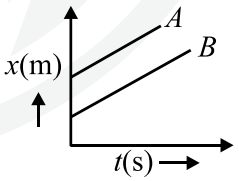
(4) 0.02 mm

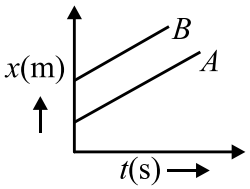
23. The sound intensity level at a distance of 10 m from a point source is 40 dB. Assuming the source emits sound equally in all directions, what is the sound intensity level at a distance of 1 m from the same source?

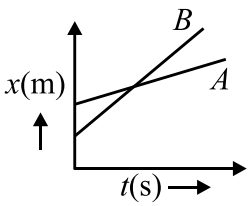
(1) 50 dB (2) 80 dB

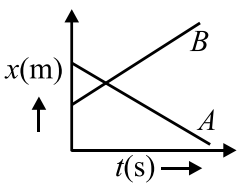
(3) 20 dB (4) 60 dB

24. Two objects A and B are moving with velocities v_A and v_B respectively along positive x-axis. If $v_A < v_B$ then, which of the following position-time graphs correctly represent the motion?

(1)


(2)


(3)


(4)


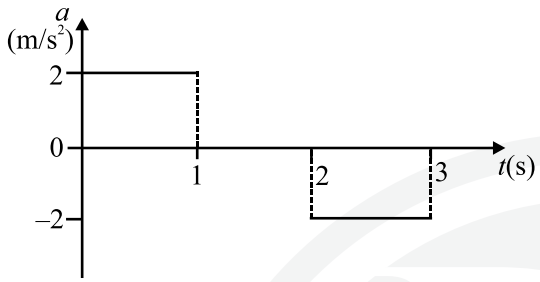
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25. A force vector of magnitude 10 N is applied at an angle of 60° measuring anticlockwise with the positive *x*-axis. Its rectangular components along the *x* and *y* axes (both in newton) respectively are:

(1) $\left(5\sqrt{3},5\right)$ (2) $\left(5,5\right)$

(3) $\left(5\sqrt{2},5\sqrt{2}\right)$ (4) $\left(5,5\sqrt{3}\right)$

26. Acceleration-time graph for a particle is given in figure. If it starts from rest at $t=0$, distance travelled in 3 s will be:



(1) 4 m (2) 2 m

(3) 0 (4) 6 m

27. The equation of a standing wave in a string is given by $y=4\sin\left(\frac{\pi x}{4}\right)\cos\left(50\pi t\right)$, where *x* and *y* are in meters. The separation between two consecutive nodes is:

(1) 8 m (2) 4 m

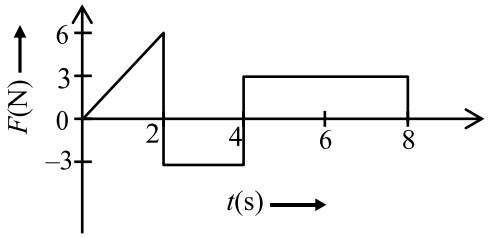
(3) 2 m (4) 1 m

28. A projectile is launched with a speed of 50 m/s at an angle of 53° with the horizontal. The magnitude of its average velocity between $t=2$ s and $t=6$ s is: (Take $g=10\ \text{m/s}^2$)

(1) $30\sqrt{2}$ m/s (2) 30 m/s

(3) 40 m/s (4) Zero

29. The force F acting on a particle of mass *m* is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from 0 s to 8 s is:



(1) 24 Ns (2) 20 Ns

(3) 12 Ns (4) 6 Ns

30. The fundamental frequency of a closed organ pipe of length 20 cm is equal to the frequency of the first overtone of an open organ pipe. The length of the open pipe is:

(1) 40 cm (2) 20 cm

(3) 80 cm (4) 60 cm

31. At a depth of 1000 m in the ocean, what is the magnitude of fractional change in the volume of a water sample? (Density of water $=10^3\ \text{kg/m}^3$, Compressibility $=5\times10^{-10}\ \text{Pa}^{-1}$, $g=10\ \text{m/s}^2$)

(1) 1.5×10^{-2} (2) 0.5×10^{-2}

(3) 1.0×10^{-3} (4) 2.0×10^{-2}

32. Internal forces acting within a system of particles can alter:

(1) The linear momentum as well as the kinetic energy of the system

(2) The linear momentum of the system, but not the kinetic energy of the system

(3) The kinetic energy of the system, but not the linear momentum of the system

(4) Neither linear momentum nor kinetic energy of the system

33. Car *A* travels East at 10 m/s. Car *B* travels North at 10 m/s. The velocity of Car *B* relative to Car *A* is:

(1) $10\sqrt{2}$ m/s North-West

(2) 10 m/s North-East

(3) $10\sqrt{2}$ m/s North-East

(4) $10\sqrt{2}$ m/s South-East

34. A monatomic gas $\gamma=\frac{5}{3}$ at pressure P and volume V is compressed isothermally to volume $\frac{V}{2}$ and then compressed adiabatically to a final volume of $\frac{V}{16}$. The final pressure of the gas is:

(1) $32P$ (2) $16P$

(3) $64P$ (4) $128P$

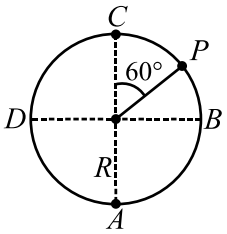
35. A theoretical material has a Poisson's ratio of 0.5. If a rod of this material is subjected to a longitudinal tensile strain of 10^{-3}, the percentage change in its volume is:

(1) 0.1% (2) 0.5%

(3) Zero (4) 0.2%

[Contd...

36. A particle starts moving in anticlockwise direction from point A moving along a vertical circle of radius R as shown. If it just completes the circle, what will be its velocity at point P? (g is acceleration due to gravity)



(1) $\sqrt{gR}$ (2) $\sqrt{2gR}$

(3) $\sqrt{3gR}$ (4) $\sqrt{\frac{3}{2}gR}$

37. A drone is flying horizontally at constant speed 100 m/s. A projectile is launched from the ground at the instant when the drone is directly overhead and at a reachable height. If the projectile speed is 200 m/s, at what angle θ with the horizontal must it be fired to hit the drone?

(1) 30° (2) 45°

(3) 60° (4) 75°

38. Viscous drag force on a spherical body moving through a viscous fluid depends on:

(1) Size of body

(2) Velocity with which it moves

(3) Viscosity of fluid

(4) All of these

39. Water flows through a horizontal main pipe of radius R with speed v. This pipe divides into n smaller identical pipes, each of radius r. The speed of water in the smaller pipes is:

(1) $\frac{vR}{nr}$ (2) $\frac{vR^2}{nr^2}$

(3) $\frac{vR^2}{n^2r^2}$ (4) $\frac{vR}{n^2r}$

40. A particle executes simple harmonic motion governed by the differential equation $2\frac{d^2x}{dt^2}+50x=0$. The time period of oscillation is: (x is in metre and t is in second)

(1) $\frac{2\pi}{5}$ s (2) $\frac{\pi}{5}$ s

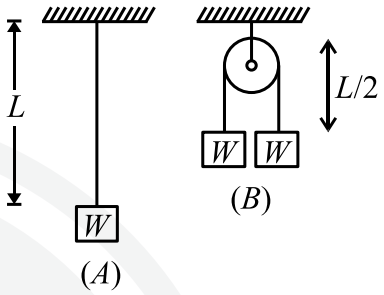
(3) 5π s (4) 2π s

41. 50 g of water at 20°C is poured onto a large block of ice at 0°C. Assuming no heat loss to surroundings, the mass of ice that melts is: (Latent heat of fusion $L=80$ cal/g)

(1) 12.5 g (2) 50 g

(3) 25 g (4) 100 g

42. A wire holds certain weights in two different ways as shown. If in case A, elongation in wire of length L is l, then for the same wire, elongation in case B will be:



(1) $4l$ (2) $2l$

(3) l (4) $\frac{l}{2}$

43. A liquid drop has a radius of 2 mm. If the surface tension of the liquid is 0.07 N/m, the excess pressure inside the drop compared to the outside is:

(1) 35 Pa (2) 140 Pa

(3) 70 Pa (4) 280 Pa

44. At a same temperature T, what is the ratio of the average translational kinetic energy per molecule of Hydrogen gas (H_2) to that of Argon gas (Ar)?

(1) 1:20 (2) 1:1

(3) 2:1 (4) 1:4

45. A cylindrical rod conducts heat along the length at a rate Q. It is melted down and recast into a new rod with half the original radius (keeping volume constant). The new rate of heat conduction across the same temperature difference is:

(1) $\frac{Q}{16}$

(2) $16Q$

(3) $\frac{Q}{4}$

(4) $4Q$

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46. For the given hypothetical reactions, the equilibrium constants are as follows:

$A \rightleftharpoons B$; $K_1 = 2.0$

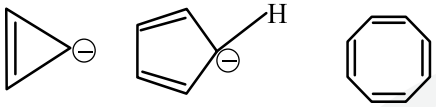
$B \rightleftharpoons C$; $K_2 = 3.0$

$C \rightleftharpoons D$; $K_3 = 2.0$

The equilibrium constant for the reaction $A \rightleftharpoons D$ is:

(1) 6.0 (2) 12.0
(3) 7.0 (4) 4.0

47. The **correct** stability order of the following species/molecules is:



p q r

(1) p > q > r (2) q > p > r
(3) q > r > p (4) r > q > p

48. Consider the following statements regarding the photoelectric effect:

1. The kinetic energy of emitted electrons depends on the intensity of incident radiation.
2. The number of emitted photoelectrons increases with increase in intensity of incident light (above threshold frequency).
3. There exists a minimum frequency below which no photoelectrons are emitted, irrespective of intensity.
4. The stopping potential is independent of the frequency of incident radiation.

Which of the above statements are **correct**?

(1) 2 and 3
(2) 1 and 4
(3) 1, 2 and 3
(4) 2, 3 and 4

49. Match the **List-I** with **List-II**.

List-I Molecule/ion		**List-II Bond pair (σ + π): lone pair (On the central atom)**	
(A)	ICl_2^-	(I)	4 : 2
(B)	H_2O	(II)	4 : 1
(C)	SO_2	(III)	2 : 3
(D)	XeF_4	(IV)	2 : 2

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-II, D-I
(2) A-III, B-IV, C-II, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-I, C-IV, D-III

50. Mass of calcium required to produce 112 mL of hydrogen gas at STP on reaction with excess dilute HCl is

Given: Molar mass of Ca = 40 g mol^{-1}.

(1) 0.20 g (2) 0.10 g
(3) 0.40 g (4) 0.80 g

51. Given below are two statements:

Statement (I): According to Newlands' law of octaves, elements were arranged in the increasing order of their atomic masses.

Statement (II): Mendeleev observed periodic repetition of chemical properties when elements were arranged in order of increasing atomic mass.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are true
(2) Statement I is false but Statement II is true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

52. One mole of an ideal gas expands isothermally and reversibly from 10 dm^3 to 20 dm^3 at 300 K. ΔU q and work done in the process respectively are:

Given: R = 8.3 JK^{-1} mol^{-1}
ln 10 = 2.3
log 2 = 0.30
log 3 = 0.48

(1) 0, 21.84 kJ, –1.726 kJ
(2) 0, –17.18 kJ, 1.718 kJ
(3) 0, 21.84 kJ, 21.84 kJ
(4) 0, 1.718 kJ, –1.718 kJ

53. Bromine undergoes disproportionation in cold alkaline medium as shown below:

$$aBr_2(g) + bOH^-(aq) \rightarrow BrO^-(aq) + dBr^-(aq) + eH_2O(l)$$

The values of a, b, c and d in the balanced redox reaction are respectively:

(1) 1, 2, 1 and 1
(2) 2, 2, 1 and 3
(3) 3, 4, 4 and 2
(4) 1, 4, 1 and 3

[Contd...

54. Consider the following statements:

(I) The pH of a mixture containing 400 mL of 0.1 M H_2SO_4 and 400 mL of 0.1 M NaOH will be approximately 1.3.

(II) Ionic product of water is temperature dependent.

(III) A monobasic acid with $K_a = 10^{-5}$ has a pH = 5. The degree of dissociation of this acid is 50%.

(IV) The Le-Chatelier's principle is not applicable to common-ion effect.

The **correct** statement are:

(1) (I), (II) and (IV)

(2) (I), (II) and (III)

(3) (II) and (III)

(4) (I), and (II)

55. Which one of the following complexes is violet in colour?

(1) $[Fe(CN)_5NOS]^{4-}$

(2) $[Fe(SCN)]^{2+}$

(3) $Fe_4[Fe(CN)_6]_3.H_2O$

(4) $[Fe(CN)_6]^{4-}$

56. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): SnF_4 and PbF_4 are covalent while $SnCl_4$ is ionic.

Reason (R): Fluorine forms strong covalent bonds due to high electronegative and small size.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

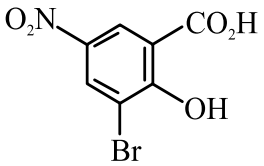
(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

57. What is the **correct** IUPAC name of



(1) 3-Bromo-2-hydroxy-5-nitrobenzoic acid

(2) 3-Bromo-4-hydroxy-1-nitrobenzoic acid

(3) 2-Hydroxy-3-bromo-5-nitrobenzoic acid

(4) 5-Nitro-3-bromo-2-hydroxybenzoic acid

58. Among the following molecules/ions, which one is diamagnetic and has the highest bond order?

(1) B_2

(2) C_2

(3) O_2

(4) O_2^-

59. Match **List-I** with **List-II.**

List I (Pair of compounds)		**List II (Type of Isomerism)**	
(A)	Propanal and Acetone	(I)	Functional isomerism
(B)	Pentane and 2-methylbutane	(II)	Chain isomerism
(C)	1-Chloropropane and 2-Chloropropane	(III)	Position isomerism
(D)	Ethoxy propane and Methoxy butane	(IV)	Metamerism

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-IV, B-II, C-I, D-III

(4) A-II, B-IV, C-III, D-I

60. Given below are two statements.

Statement-I: Spectral lines are produced when an electron absorbs energy and jumps to a higher orbit only.

Statement-II: Emission spectra are obtained when electrons transition from higher to lower energy levels.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.

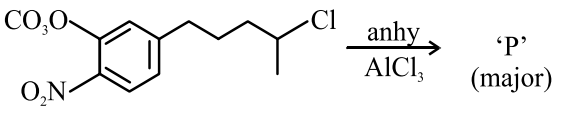
(2) Both Statement I and Statement II are incorrect.

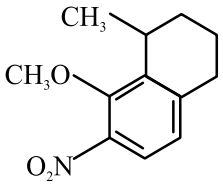
(3) Statement I is correct but Statement II is incorrect.

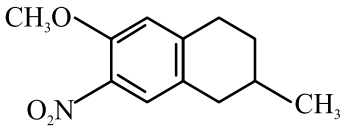
(4) Statement I is incorrect but Statement II is correct.

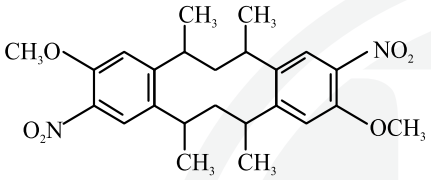
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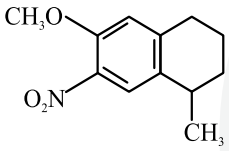
61. The major product 'P' formed in the given reaction is:



(1) 

(2) 

(3) 

(4) 

62. Consider the following statements:

A. Hydronium ion in aqueous solution is further solvated.

B. Species like $H_5O_2^+$ and $H_9O_4^+$ can exist due to hydrogen bonding.

C. Hydration decreases the stability of hydronium ion.

D. Hydrated hydronium ions are stabilized by intermolecular hydrogen bonding.

Which of the above statements are **correct**?

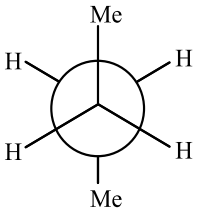
(1) A, B and D

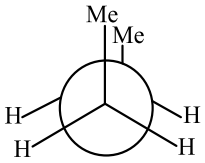
(2) B and C

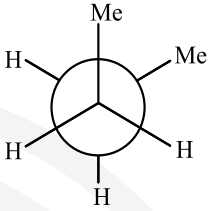
(3) A and C

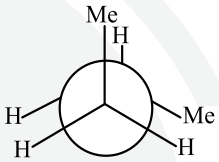
(4) A, B and C

63. Which of following configuration will be the most stable?

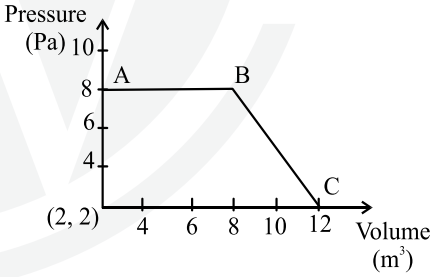
(1) 

(2) 

(3) 

(4) 

64. The magnitude of work done by a gas that undergoes a reversible expansion along the path ABC shown in the figure is



(1) 48 J

(2) 38 J

(3) 28 J

(4) 58 J

65. A weak acid HA has degree of dissociation *x*. Which option gives the **correct** expression of (pH – pKa)?

(1) $\log(1+2x)$

(2) 0

(3) $\log\left(\frac{x}{1-x}\right)$

(4) $\log\left(\frac{1-x}{x}\right)$

[Contd...

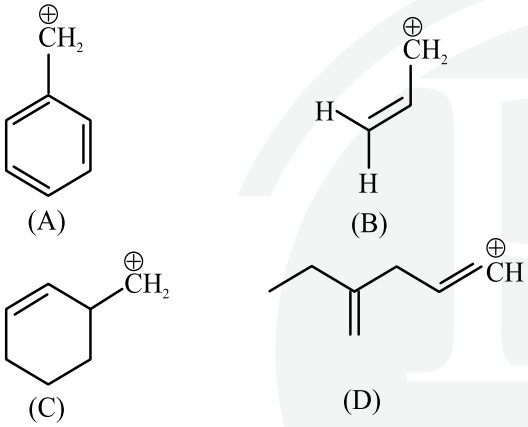
66. Match **List I** with **List II.**

List I (Element)		List II (Ionic Radius in pm)	
(A)	Sn^{4+}	(I)	40
(B)	Si^{4+}	(II)	69
(C)	Pb^{4+}	(III)	78
(D)	Ge^{4+}	(IV)	53

Thus, the **correct** matching is:

(1) A–II, B–I, C–III, D–IV
(2) A–I, B–IV, C–II, D–III
(3) A–II, B–IV, C–I, D–III
(4) A–III, B–II, C–IV, D–I

67. Among the given species the Resonance stabilized carbocations are:



(1) (A), (B) and (C) only
(2) (C) and (D) only
(3) (A), (B) and (D) only
(4) (A) and (B) only

68. Which of the following statements are **correct**?

A. The process of adding an electron to a neutral gaseous atom is always exothermic.
B. The process of removing an electron from an isolated gaseous atom is always endothermic.
C. The first ionization enthalpy of oxygen is less than that of nitrogen.
D. The electronegativity of an element is the same in all its compounds.
E. Sodium is the most electropositive element among alkali metals.

Choose the **correct** answer from the options given below:

(1) B and C only
(2) A, C and D only
(3) A and C only
(4) B, C and E only

69. The average atomic mass of an element X is 24.5 u. The element has two isotopes ^{24}X and ^{25}X. The ratio of ^{24}X to ^{25}X in naturally occurring sample is closest to:

(1) 1 : 1 (2) 2 : 1
(3) 3 : 1 (4) 1 : 3

70. Which of the following options are **correct** for the reaction

$2[Au(CN)_2]^-_{(aq)} + Zn(s) \rightarrow 2Au(s) + [Zn(CN)_4]^{-2}_{(aq)}$?

A. Redox reaction
B. Displacement reaction
C. Decomposition reaction
D. Combination reaction

Choose the **correct** answer from the options given below:

(1) A and B only (2) A only
(3) C and D only (4) A and D only

71. Given below are two statements:

Statement (I): Neopentane gives only one monochloro derivative on free radical chlorination.

Statement (II): Boiling point of neopentane is lower than that of n-pentane.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Statement I is correct but Statement II is incorrect
(2) Both Statement I and Statement II are correct
(3) Both Statement I and Statement II are incorrect
(4) Statement I is incorrect but Statement II is correct

72. Given below are two statements:

Statement-I: IUPAC name of

$HO - CH_2 - (CH_2)_3 - CH_2 - COCH_3$

is 7-hydroxyheptan-2-one.

Statement-II: 2-oxoheptan-7-ol is the correct IUPAC name for above compound.

In the light of the above statements. Choose the most appropriate answer from the options given below.

(1) Statement-I is correct but Statement-II is incorrect.
(2) Both Statement-I and Statement-II are incorrect.
(3) Both Statement-I and Statement-II are correct.
(4) Statement-I is incorrect but Statement-II is correct.

73. Electronic configurations of four elements P, Q, R and S are given below:

P. $1s^2\ 2s^2\ 2p^1$ Q. $1s^2\ 2s^2\ 2p^4$

R. $1s^2\ 2s^2\ 2p^5$ S. $1s^2\ 2s^2\ 2p^2$

Which of the following is the **correct** order of increasing electronegativity (Pauling scale)?

(1) P < S < Q < R (2) S < P < Q < R

(3) P < Q < S < R (4) S < Q < P < R

74. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The dipole moment of NF_3 is smaller than that of NH_3 even though fluorine is more electronegative than hydrogen.

Reason (R): In NF_3, the resultant bond dipole moments are directed opposite to the dipole moment due to the lone pair on nitrogen, leading to partial cancellation.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

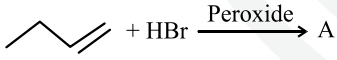
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

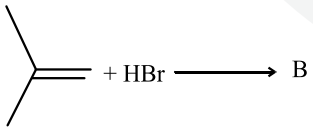
(2) (A) is false but (R) is true

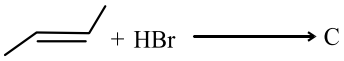
(3) (A) is true but (R) is false

(4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

75. The increasing order of the boiling points of the major products A, B and C of the following reactions will be:

A. 

B. 

C. 

(1) B < C < A (2) C < A < B

(3) A < B < C (4) A < C < B

76. For a concentrated solution of a weak electrolyte (Keq = equilibrium constant)

A_2B_3 of concentration 'c', the degree of dissociation 'α' is

(1) $\left(\frac{K_{eq}}{108c^4}\right)^{\frac{1}{5}}$ (2) $\left(\frac{K_{eq}}{6c^5}\right)^{\frac{1}{5}}$

(3) $\left(\frac{K_{eq}}{5c^4}\right)^{\frac{1}{5}}$ (4) $\left(\frac{K_{eq}}{25c^2}\right)^{\frac{1}{5}}$

77. The Lewis acid character of boron tri halides follows the order:

(1) $BBr_3 > BI_3 > BCl_3 > BF_3$

(2) $BCl_3 > BF_3 > BBr_3 > BI_3$

(3) $BF_3 > BCl_3 > BBr_3 > BI_3$

(4) $BI_3 > BBr_3 > BCl_3 > BF_3$

78. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): The first ionisation enthalpy generally increases across a period.

Reason (R): The effective nuclear charge increases across a period as shielding remains nearly constant.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true but R is NOT the correct explanation of A

79. In order to prepare a buffer solution of pH 5.74, sodium acetate is added to acetic acid. If the concentration of acetic acid in the buffer is 1.0 M, the concentration of sodium acetate in the buffer is _____.

[Given: pKa (acetic acid) = 4.74]

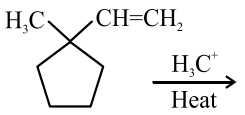
(1) 10 molar

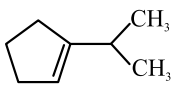
(2) 10^{-1} molar

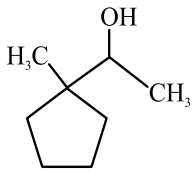
(3) 100 molar

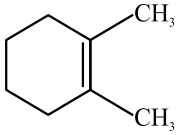
(4) 10^{-2} molar

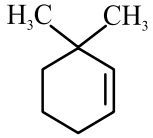
80. The major product in the following reaction is:



(1) 

(2) 

(3) 

(4) 

[Contd...

81. How many orbitals in an atom can have the quantum numbers n = 4 and l = 3?

(1) 5 (2) 3

(3) 7 (4) 9

82. At 300 K and 1 atmospheric pressure, 10 mL of a gaseous hydrocarbon requires 60 mL of O_2 for complete combustion and 40 mL of CO_2 is formed. The molecular formula of the hydrocarbon is:

(1) C_4H_{10} (2) C_4H_8

(3) C_4H_6 (4) C_3H_8

83. For the reaction $C_2H_6 \rightarrow C_2H_4 + H_2$, the reaction enthalpy Δ_rH = _____ kJ mol^{-1}. (Round off to the nearest integer).

[Given: Bond enthalpies in kJ mol^{-1}; C = C: 347, C = C : 611, C – H : 414, H–H : 436];

(1) 128 (2) 138

(3) 158 (4) 178

84. Which of the following **cannot** function as an oxidising agent?

(1) N^{3-} (2) SO_4^{2-}

(3) BrO_3^- (4) BrO_4^-

85. Identify product [A], [B] and [C] in the following reaction sequence.

$$CH_3-C\equiv CH \xrightarrow[H_2]{Pd/C} [A] \xrightarrow[(ii)Zn,H_2O]{(i)O_3} [B]+[C]$$

(1) [A]:$CH_3CH_2CH_3$, [B]: CH_3CHO, [C]: HCHO

(2) [A]:$CH_2=CH_2$, [B]: $H_3C–CO–CH_3$, [C]: HCHO

(3) [A]:$CH_3–CH=CH_2$, [B]:CH_3CHO, [C]: CH_3CH_2OH

(4) [A]:$CH_3–CH = CH_2$, [B]: CH_3CHO, [C]: HCHO

86. At –20°C and 1 atm pressure, a cylinder is filled with equal number of mole of H_2, I_2 and HI for the reaction

$H_2(g)+I_2(g) \rightleftharpoons 2HI(g)$, the Kp for the process is $x \times 10^{-1}$.

The value of x is?

[Given: R = 0.082 L atm K^{-1} mol^{-1}]

(1) 2

(2) 1

(3) 10

(4) 0.01

87. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The cis form of 2-butene is more polar than the trans form.

Reason (R): The dipole moments of the two $C–CH_3$ bonds in trans-2-butene cancel each other due to symmetrical arrangement.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

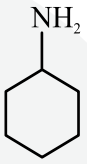
(1) Both A and R are true and R is the correct explanation of A.

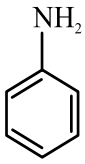
(2) A is true but R is false.

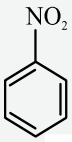
(3) A is false but R is true.

(4) Both A and R are true but R is NOT the correct explanation of A.

88. The Kjeldahl's method of Nitrogen estimation fails for which of the following ?

(A) 

(B) 

(C) 

(D) 

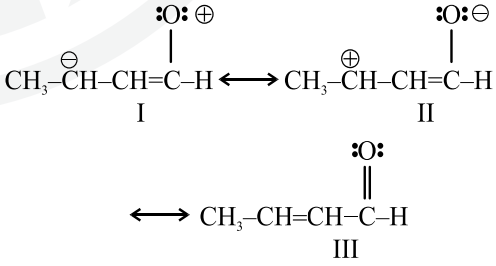
(1) (C) and (D)

(2) (B) and (C)

(3) (A) and (D)

(4) (A), (C) and (D)

89. The **correct** stability order of following resonance structures of $CH_3 – CH = CH – CHO$ is



(1) II > III > I (2) III > II > I

(3) I > II > III (4) II > I > III

90. For emission line of atomic hydrogen from $n_1 = 8$ to $n_f = n$, the plot of wave number $(\bar{\nu})$ against $\left(\frac{1}{n^2}\right)$ will be (The Rydberg constant R_H is in wave number unit):

(1) Linear with intercept - R_H

(2) Non linear

(3) Linear with slope R_H

(4) Linear with slope - R_H

[Contd...

91. In taxonomy, families are characterized on the basis of which features of plant species?
(1) Vegetative features only
(2) Reproductive features only
(3) Both vegetative and reproductive features
(4) Anatomical features only

92. Which scientist concluded that the presence of a cell wall is a unique character of plant cells, based on his studies on plant tissues?
(1) Matthias Schleiden
(2) Theodore Schwann
(3) Rudolf Virchow
(4) Antonie Von Leeuwenhoek

93. Match **List-I** with **List-II**.

List-I (Plant Hormone)		**List-II (Physiological effect)**	
(A)	Auxin	(I)	Adventitious shoot growth
(B)	Cytokinin	(II)	Promotes rapid internode/petiole elongation in deep-water rice plant
(C)	Gibberellins	(III)	Improves shape of apple fruit
(D)	Ethylene	(IV)	Xylem differentiation

Choose the **correct** answer from the options given below:
(1) A–IV, B–I, C–II, D–III
(2) A–III, B–I, C–II, D–IV
(3) A–IV, B–I, C–III, D–II
(4) A–I, B–IV, C–II, D–III

94. Identify the pair of phytohormones that are naturally occurring in plants.
(1) GA_3 and Kinetin
(2) IBA and Kinetin
(3) Zeatin and IBA
(4) ABA and 2, 4-D

95. In members of Chlorophyceae, the storage bodies located in the chloroplasts are called:
(1) pyrenoids.
(2) mannitol.
(3) floridean starch.
(4) laminarin.

96. Which of the following is **not** a plant factor that affects photosynthetic rate?
(1) Temperature
(2) Orientation of leaves
(3) Internal CO_2 concentration
(4) Amount of chlorophyll

97. The tail of the lipid molecule in the plasma membrane is:
(1) hydrophilic and toward the outer side.
(2) hydrophobic and toward the inner side.
(3) hydrophilic and toward the inner side.
(4) hydrophobic and toward the outer side.

98. Given below are two statements:
Statement I: In diatoms, the cell walls form two thin overlapping shells which fit together as in a soap box.
Statement II: Diatoms are microscopic and float actively in water currents due to the presence of flagella.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

99. In the coconut fruit, the mesocarp is:
(1) fleshy and edible.
(2) stony and hard.
(3) fibrous.
(4) thin and membranous.

100. The stage between the two meiotic divisions is:
(1) long lived with DNA replication.
(2) generally short lived and has no DNA replication.
(3) short lived with DNA replication.
(4) long lived without DNA replication.

101. Aestivation in which the margins of sepals or petals overlap one another but not in any particular direction is seen in:
(1) pea. (2) cotton.
(3) gulmohur. (4) *Calotropis.*

102. Which of the following statements regarding the "Two Kingdom" classification system is **incorrect**?
(1) It divided organisms into Plantae and Animalia.
(2) It distinguished between eukaryotes and prokaryotes.
(3) It was used for a long time but found inadequate.
(4) It included bacteria and fungi under plants.

103. Identify the **incorrect** statement regarding mesosomes.
(1) These are the extensions of the plasma membrane into the cell.
(2) They help in cell wall formation and DNA replication.
(3) They form middle layer of cell envelope.
(4) They are present in the form of vesicles, tubules and lamellae.

104. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The spread of living pteridophytes is limited and restricted to narrow geographical regions.
Reason (R): The sporophyte require cool, damp, shady places to grow and water is needed for fertilization.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

105. The reaction centre chlorophyll *a* in PS I and PS II shows absorption maxima at, respectively:
(1) 680 nm and 700 nm.
(2) 700 nm and 680 nm.
(3) 680 nm and 660 nm.
(4) 720 nm and 700 nm.

106. Match **List-I** with **List-II.**

List I (Cell Type)		**List II (Description)**	
(A)	Tracheids	(I)	Perforated end walls, lacks nucleus
(B)	Vessels	(II)	Elongated, tapering ends, dead
(C)	Sieve tube elements	(III)	Cylindrical tube, multicellular, lignified
(D)	Companion cells	(IV)	Specialized parenchyma, helps in maintaining pressure gradient

Choose the **correct** answer from the options given below:
(1) A–II, B–III, C–I, D–IV
(2) A–III, B–II, C–I, D–IV
(3) A–II, B–I, C–III, D–IV
(4) A–I, B–II, C–IV, D–III

107. Match **List-I** with **List-II.**

List-I (Structure)		**List-II (Function/Location)**	
(A)	Heterocyst	(I)	Motility in bacteria
(B)	Pellicle	(II)	Nitrogen fixation
(C)	Flagellum	(III)	Protein coat of virus
(D)	Capsid	(IV)	Flexibility in Euglenoids

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-II, C-I, D-III
(4) A-II, B-IV, C-III, D-I

108. Which of the following is a characteristic that is found only in human being?
(1) Reproduction
(2) Metabolism
(3) Cellular organisation
(4) Self-consciousness

109. Given below are two statements.
Statement I: In gymnosperms, released pollen grain are carried in water currents.
Statement II: In conifers, the needle-like leaves, thick cuticle and sunken stomata help to reduce water loss.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

110. In the photorespiratory pathway, there is:
(1) synthesis of sugars and ATP.
(2) synthesis of ATP but not NADPH.
(3) release of CO_2 with utilization of ATP.
(4) no release of CO_2 but synthesis of ATP.

111. Agar is obtained from:
(1) *Laminaria* and *Sargassum.*
(2) *Gelidium* and *Gracilaria.*
(3) *Chlorella* and *Porphyra.*
(4) *Volvox* and *Ulothrix.*

112. The final stage of meiotic prophase I is marked by:
(1) terminalisation of chiasmata.
(2) formation of chiasmata.
(3) dissolution of synaptonemal complex.
(4) appearance of recombination nodules.

[Contd...

113. Which of the following statements about auxins are **true**?
A. First isolated from autoclaved herring sperm DNA.
B. Promote flowering in pineapples.
C. Help to prevent fruit and leaf drop at early stages.
D. Promote the abscission of older mature leaves and fruits.
E. Help to overcome apical dominance.
Choose the **correct** answer from the options given below:
(1) A, B and D only (2) A and E only
(3) B, C and D only (4) A, B, C and E only

114. Which of the following sets of plants possess apocarpous gynoecium?
(1) Lotus and Rose
(2) Mustard and Tomato
(3) Pea and China rose
(4) *Datura* and *Petunia*

115. The prothallus of pteridophytes is:
(1) conspicuous, large, and diploid.
(2) inconspicuous, small, free-living, and haploid.
(3) parasitic and non-photosynthetic.
(4) the dominant sporophyte stage.

116. Identify the **correct** characteristic feature of the monocot stem anatomy.
(1) Vascular bundles are arranged in a ring.
(2) Vascular bundles are conjoint and open.
(3) Phloem parenchyma is absent.
(4) Pith is large and well-developed.

117. The final hydrogen acceptor in aerobic respiration is:
(1) cytochrome *c*. (2) oxygen.
(3) NAD^+. (4) water.

118. The main criteria for classification used by R.H. Whittaker included all of the following **except**:
(1) Cell structure and body organization
(2) Mode of nutrition
(3) Morphological characters
(4) Phylogenetic relationships

119. Which of the following statements about the Golgi apparatus is true?
(1) The *cis* and the *trans* faces of the organelle are entirely identical.
(2) Materials to be packaged fuse with the *trans* face and exit from the *cis* face.
(3) It is the important site of formation of glycoproteins and glycolipids.
(4) It was first observed by George Palade in 1898.

120. Match **List I** with **List II**.

List-I (Organism)		**List-II (Structure/Feature)**	
(A)	*Marchantia*	(I)	Prothallus
(B)	*Cycas*	(II)	Gemma cup
(C)	Fern	(III)	Coralloid roots
(D)	*Sphagnum*	(IV)	Peat formation

Choose the **correct** answer from the options given below:
(1) A–II, B–IV, C–I, D–III
(2) A–I, B–IV, C–II, D–III
(3) A–II, B–III, C–I, D–IV
(4) A–III, B–II, C–IV, D–I

121. The main arena of various cellular activities in both plant and animal cells is:
(1) nucleus.
(2) cytoplasm.
(3) plasma membrane.
(4) mitochondria.

122. During non-cyclic photo-phosphorylation, the splitting of water is associated with:
(1) PS I on the outer side of the thylakoid membrane.
(2) PS II on the outer side of the thylakoid membrane.
(3) PS II on the inner side of the thylakoid membrane.
(4) PS I on the inner side of the thylakoid membrane.

123. Identify the **correct** statements regarding Ascomycetes.
A. The mycelium is branched and septate.
B. Asexual spores are conidia produced endogenously.
C. Sexual spores are ascospores produced endogenously in asci.
D. *Neurospora* is used in biochemical and genetic work.
E. Shows dikaryon phase in sexual reproduction.
Choose the **correct** answer from the options given below:
(1) A, B, D and E
(2) A, C, D and E
(3) B, C and D
(4) A and D only

124. Given below are two statements.
Statement I: Growth at a cellular level is principally a consequence of an increase in the amount of protoplasm.
Statement II: ABA plays an important role in seed development, maturation and dormancy.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

125. Flower cannot be divided into two similar halves by any vertical plane passing through the centre in:
(1) canna. (2) *Datura.*
(3) *Cassia.* (4) Chilli.

126. In Meiosis I, the homologous chromosomes separate, while sister chromatids remain associated at their centromeres during:
(1) anaphase I. (2) anaphase II.
(3) metaphase I. (4) metaphase II

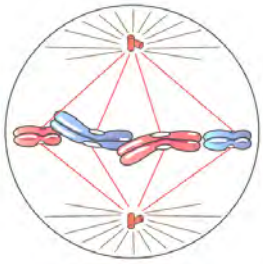
127. Consider the following statements about Gymnosperms.
A. The roots are generally tap roots
B. All gymnosperms are homosporous
C. The male and female gametophytes do not have an independent existence
D. They possess tracheids, vessels and albuminous cells
E. The lateral meristems appear later in life
How many of the above statements are **correct**?
(1) One only (2) Two only
(3) Three only (4) Four only

128. Which of the following is **not** a feature of plants that allows them to survive without respiratory organs?
(1) Each plant part takes care of its own gas exchange needs.
(2) There is very little transport of gases from one plant part to another.
(3) Loose packing of parenchyma cells provides interconnected air spaces.
(4) Plants have a high demand for gas exchange during respiration.

129. The tangential and radial walls of endodermal cells in dicot root have a deposition of:
(1) lignin. (2) suberin.
(3) pectin. (4) cellulose.

130. *Solanum*, *Petunia*, and *Datura* are placed in which Order?
(1) Polymoniales (2) Sapindales
(3) Poales (4) Solanaceae

131. Identify the **correct** feature of the stage of cell cycle given below.



(1) Spindle fibres attach to the centromere region.
(2) Chromosomes are present inside the nucleus.
(3) Chromosomes appear as bivalents.
(4) By this stage, condensation of chromosomes is complete.

132. Which of the following is **not** a modification of the stem?
(1) Pneumatophores in *Rhizophora*
(2) Thorns in *Citrus*
(3) Tendrils in cucumber
(4) Fleshy cylindrical structure in *Euphorbia*

133. F_1 particle of ATP synthase:
(1) is an integral membrane protein complex.
(2) forms the channel through which protons pass.
(3) is a peripheral membrane protein complex containing the catalytic site.
(4) is located towards intermembrane space of mitochondria.

134. George Bentham and Joseph Dalton Hooker are associated with which type of classification system?
(1) Artificial system
(2) Phylogenetic system
(3) Numerical Taxonomy
(4) Natural system

135. Which organelle is responsible for the synthesis of the steroidal hormones in animal cells?
(1) Rough Endoplasmic Reticulum
(2) Smooth Endoplasmic Reticulum
(3) Golgi apparatus
(4) Ribosomes

136. In cockroach, the seminal vesicles primarily function to:
(1) produce spermatophores.
(2) store sperms before copulation.
(3) secrete accessory gland fluid.
(4) transfer sperms directly to the oviduct.

137. What is the typical volume of functional residual capacity (FRC) in an adult human being?
(1) 3000 ml (2) 2300 ml
(3) 1500 ml (4) 1100 ml

138. If the tertiary structure of an enzyme is disrupted, the most immediate consequence is:
(1) increase in peptide bond formation.
(2) conversion into nucleic acid.
(3) loss of biological activity.
(4) formation of additional α-helices.

139. Passage of ova through the female reproductive tract is facilitated by:
(1) Muscular movement only
(2) Flagellar movement
(3) Amoeboid movement
(4) Ciliary movement

140. Cretinism in children is caused by which of the following conditions in the mother during pregnancy?
(1) Hypothyroidism
(2) Hyperthyroidism
(3) Hypoparathyroidism
(4) Hyperparathyroidism

141. Given below are two statements:
Statement I: Ligament is a dense irregular connective tissue.
Statement II: Cartilage is a dense regular connective tissue.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct, but Statement II is incorrect.
(3) Statement I is incorrect, but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

142. The limbic system is formed by;
(1) hypothalamus, epithalamus, amygdala and pons.
(2) the inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus etc.
(3) corpora quadrigemina and hippocampus.
(4) midbrain and hindbrain.

143. Which type of epithelium is present in the air sacs of the lungs?
(1) Simple squamous epithelium
(2) Simple columnar epithelium
(3) Simple cuboidal epithelium
(4) Compound epithelium

144. Which of the following factors can increase the sodium reabsorption in the DCT?
(1) Decrease in aldosterone level.
(2) Decrease in vasopressin level.
(3) Increase in aldosterone level.
(4) Increase in vasopressin levels.

145. In a normal ECG, which of the following represents the depolarization of the ventricles?
(1) P-wave
(2) QRS complex
(3) T-wave
(4) S-T segment

146. All of the following are coelomates exhibiting three germ cell layers, **except**;
(1) *Aplysia*
(2) *Aurelia*
(3) *Aptenodytes*
(4) *Anopheles*

147. Which of the following characteristics is **not** typical of white muscle fibers?
(1) Fewer mitochondria
(2) Anaerobic energy production
(3) High levels of sarcoplasmic reticulum
(4) High myoglobin content

148. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Platyhelminthes	(I)	Indirect development and stomochord
(B)	Annelida	(II)	Ostia and spongocoel
(C)	Hemichordata	(III)	Longitudinal and circular muscles
(D)	Porifera	(IV)	Acoelomate and bilateral symmetry

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-III, C-I, D-IV

149. Select the **incorrect** match.
(1) *Hydra* → Fresh water polyp
(2) *Branchiostoma* → Protochordate
(3) *Petromyzon* → Jawless invertebrate
(4) *Pristis* → 2-chambered heart

150. Why does the cerebellum have a very convoluted surface?
(1) To provide additional space for memory.
(2) To provide the additional space for many more neurons.
(3) For critical thinking.
(4) To control respiration and cardiovascular reflexes.

151. Which of the following is a secondary messenger in hormone action?
(1) cAMP (2) ATP
(3) ADP (4) DNA

152. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Inspiratory capacity	(I)	Total volume of air a person can inspire after normal expiration
(B)	Expiratory capacity	(II)	The maximum volume of air a person can breathe in after a forced expiration
(C)	Functional residual capacity	(III)	Volume of air that will remain in the lungs after a normal expiration
(D)	Vital capacity	(IV)	Total volume of air a person can expire after a normal inspiration

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-I, B-IV, C-III, D-II
(3) A-I, B-IV, C-II, D-III
(4) A-IV, B-I, C-III, D-II

153. The outline of principal events of urination is given below in a random manner.
A. Stretch receptors on the wall of the urinary bladder send signals to the CNS.
B. The bladder fills with urine and becomes stretched.
C. Release of urine.
D. CNS passes on motor messages to initiate the contraction of smooth muscles of the bladder and simultaneous relaxation of the urethral sphincter.
The **correct** sequence of the events is;
(1) A → B → C → D
(2) D → C →B → A
(3) B → A → D → C
(4) C →B → A → D

154. The ionic gradients across the resting membrane are maintained by the sodium-potassium pump which transport;
(1) $2Na^+$ inwards and $3K^+$ out of the cell.
(2) $3Na^+$ outwards and $2K^+$ into the cell.
(3) $2Na^+$ outwards and $3K^+$ into the cell.
(4) $3Na^+$ inwards and $2K^+$ out of the cell.

155. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Eosinophils	(I)	Immune response
(B)	Basophils	(II)	Phagocytosis
(C)	Neutrophils	(III)	Resist infections and associated with allergic reaction
(D)	Lymphocytes	(IV)	Release histamine

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-I, C-II, D-III
(3) A-I, B-II, C-IV, D-III
(4) A-II, B-I, C-III, D-IV

156. Which of the following **correctly** matches the partial pressure of CO_2 in venous blood and alveolar air, respectively?
(1) 40 mm Hg and 45 mm Hg
(2) 45 mm Hg and 40 mm Hg
(3) 104 mm Hg and 40 mm Hg
(4) 95 mm Hg and 45 mm Hg

157. Which of the following **correctly** matches the protein and its structural role?
(1) Troponin → forms the backbone of thin filament
(2) Tropomyosin → masks active sites independently in resting muscle
(3) Heavy meromyosin → forms cross arms with ATPase activity
(4) Light meromyosin → binds ATP and actin

158. Given below are two statements:
Statement I: Inorganic catalysts work efficiently at high temperature and high pressures.
Statement II: Collagen is the most abundant protein in animal world.
In the light of above statements, choose the **correct** option given below:
(1) Statement I and Statement II both are correct.
(2) Statement I and Statement II both are incorrect.
(3) Statement I is correct, but Statement II is incorrect.
(4) Statement I is incorrect, but Statement II is correct.

[Contd...

159. Identify the enzyme which catalyses the formations of carbonic acid from carbon dioxide and water.
(1) Peroxidase
(2) Nicotinamide adenine dinucleotide
(3) Carboxypeptidase
(4) Carbonic anhydrase

160. Which of the following **correctly** describes the location of the kidneys?
(1) Attached to the dorsal inner wall of the thoracic cavity
(2) Attached to the dorsal inner wall of the abdominal cavity
(3) Attached to the ventral inner wall of the abdominal cavity
(4) Attached to the ventral inner wall of the thoracic cavity

161. Which of the following glands produces the anti-inflammatory hormone?
(1) Adrenal cortex (2) Adrenal medulla
(3) Parathyroid (4) Pancreas

162. A characteristic wheezing sound is produced in which of the following respiratory disorders?
(1) Emphysema (2) Asthma
(3) Silicosis (4) Fibrosis

163. Which of the following **correctly** describes the source and release of oxytocin?
(1) Synthesised and released by anterior pituitary.
(2) Synthesised in hypothalamus and released by posterior pituitary.
(3) Synthesised and released by hypothalamus.
(4) Synthesised in posterior pituitary and released by hypothalamus.

164. The association areas in the brain is **not** responsible for;
(1) intersensory associations.
(2) communication.
(3) regulation of sexual behaviour.
(4) memory.

165. Which of the following combinations of characters is **correct** for Arthropoda?
(1) Diploblastic, radial symmetry, closed circulation
(2) Triploblastic, bilateral symmetry, open circulation
(3) Triploblastic, radial symmetry, open circulation
(4) Diploblastic, bilateral symmetry, closed circulation

166. Select the **odd one out** regarding mode of excretion of nitrogenous waste.
(1) Terrestrial amphibians
(2) Aquatic amphibians
(3) Aquatic insects
(4) Bony fishes

167. If the rate of a reaction at 40°C is 8 units, what would be the approximate rate at 20°C, assuming the general principle of effect of temperature on enzymatic activity?
(1) 1 unit
(2) 2 units
(3) 4 units
(4) 16 units

168. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In frog, the medulla oblongata continues posteriorly as the spinal cord through the foramen magnum.
Reason R: The spinal cord and brain together constitute the central nervous system, which is enclosed within protective bony structures.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

169. Which of the following bones acts as the ventral midline of the thorax?
(1) Sternum (2) Ribs
(3) Hyoid bone (4) Limb bones

170. Select the **incorrect** option w.r.t. blood vessels in humans.
(1) Each artery and vein consists of three layers.
(2) Tunica intima is the inner lining composed of squamous endothelium.
(3) Tunica media is comparatively thin in the veins.
(4) Tunica externa is the outer layer composed of smooth muscles and elastic fibres.

171. The thorax of cockroach is composed of:
(1) Two segments bearing legs and wings.
(2) Three segments, each typically bearing a pair of legs.
(3) Four segments with jointed appendages.
(4) Three segments bearing antennae.

[Contd...

172. What is the **correct** order of elemental composition in the human body, listed from highest to lowest?
(1) Oxygen > Carbon > Nitrogen > Calcium
(2) Carbon > Oxygen > Nitrogen > Calcium
(3) Calcium > Carbon > Oxygen > Nitrogen
(4) Oxygen > Calcium > Carbon > Nitrogen

173. Which of the following pairs of hormones stimulates erythropoiesis?
(1) Cortisol and adrenaline
(2) Growth hormone and adrenaline
(3) Cortisol and aldosterone
(4) Cortisol and erythropoietin

174. Which of the following is **correct** for atherosclerosis?
(1) Lumen of the vein that supplies blood to the heart muscle gets wider.
(2) Obstruction in the blood vessel leads to an increase in blood flow to heart muscle.
(3) It is caused by the deposition of fats, and cholesterol in walls of the vein.
(4) Build up of calcium, fat and fibrous tissue makes the lumen of arteries narrower in coronary blood supply.

175. Given below are two statements:
Statement I: Synovial joints are characterised by the presence of a fluid filled synovial cavity between the articulating surfaces of the two bones.
Statement II: Force generated by the muscles is used to carry out movement through joints, where the joint acts as a fulcrum.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

176. Match **List-I** with **List-II**.

List-I		List-II	
(A)	20-25% of CO_2	(I)	Deliver around 4 ml of CO_2 to the alveoli
(B)	100 ml of deoxygenated blood	(II)	Transported by RBCs as carbamino-haemoglobin
(C)	3% of O_2	(III)	Deliver around 5 ml of O_2 to the tissue
(D)	100 ml of oxygenated blood	(IV)	Transported in dissolved state through the plasma

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-I, C-IV, D-III

177. Which pair is **correctly** matched?
(1) Tetany → Uric acid deposition
(2) Gout → Hypocalcemia
(3) Osteoporosis → Decreased bone mass
(4) Muscular dystrophy → Autoimmune disorder

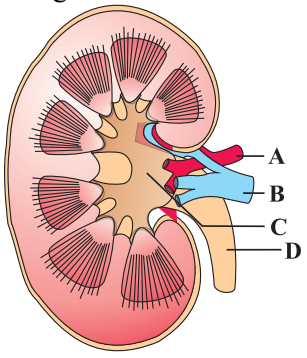
178. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Marine bony fish	(I)	Angel fish
(B)	Freshwater fish	(II)	Sea horse
(C)	Aquarium fish	(III)	Sting ray
(D)	Cartilaginous fish	(IV)	Katla

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-II, C-III, D-I
(4) A-III, B-IV, C-I, D-II

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: SA Node is the pacemaker of our heart.
Reason R: The SAN can generate the maximum number of action potentials, i.e., 70-75 min^{-1} , and is responsible for initiating and maintaining the rhythmic contractile activity of the heart.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

180. Which of the following represents the renal pelvis in the given figure?



(1) A (2) B
(3) C (4) D

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अपनी भावनाओं को समझें, तनाव और उलझनों से बाहर निकलें।

आप अकेले नहीं हैं – आपकी मदद के लिए PW प्रेरणा काउंसलर हमेशा तैयार हैं।

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